

Integration

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Section 1

Fundamental theorem of calculus

What integration does

- ◆ Integration is the continuous analogue of *summation*.
- ◆ A definite integral accumulates *signed* area; if $f \geq 0$, it is geometric area.
- ◆ In economics, integrals turn marginal objects into total objects: marginal cost into cost, densities into probabilities, and flow payoffs into present value.
- ◆ It is related to differentiation by the **Fundamental Theorem of Calculus**.

Riemann sums

Let f be **continuous** on $[a, b]$. Choose a partition

$$a = x_0 < x_1 < \cdots < x_n = b, \quad \Delta x_i = x_i - x_{i-1},$$

and a point $x_i^* \in [x_{i-1}, x_i]$ in each subinterval. The corresponding Riemann sum is $S_P = \sum_{i=1}^n f(x_i^*) \Delta x_i$, and the mesh of the partition is $\|P\| = \max_i \Delta x_i$.

Definite integral.

$$\int_a^b f(x) dx = \lim_{\|P\| \rightarrow 0} \sum_{i=1}^n f(x_i^*) \Delta x_i.$$

The integral is a limit of increasingly fine finite sums: continuous accumulation from local pieces. Continuity is sufficient but not necessary: **bounded** functions that are **continuous almost everywhere** are also Riemann integrable.

Video: [Khan Academy](#), *The definite integral as the limit of a Riemann sum*.

A numerical Riemann sum

Four right endpoint rectangles for $f(x) = x^2$

Divide $[0, 1]$ into four intervals of width $1/4$ and use $x_i^* = i/4$. Then

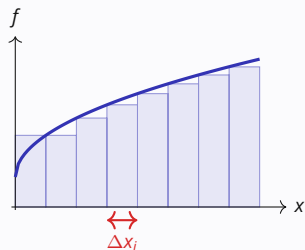
$$S_4 = \sum_{i=1}^4 \left(\frac{i}{4}\right)^2 \frac{1}{4} = \frac{1^2 + 2^2 + 3^2 + 4^2}{64} = \frac{15}{32}.$$

Since x^2 is increasing, these right endpoint rectangles lie above the graph on each subinterval. Finer partitions make the sums approach

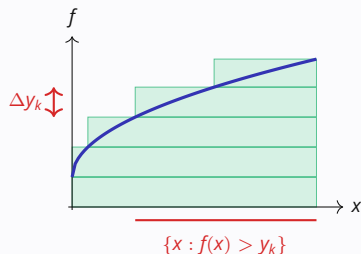
$$\int_0^1 x^2 dx = \frac{1}{3}.$$

Cutting the domain or cutting the range

The same area under f can be built up in two directions. Riemann cuts the x -axis into small intervals; Lebesgue cuts the y -axis into small bands and asks how wide the set of x with $f(x)$ above each level is.



$$\sum_i f(x_i^*) \Delta x_i \quad \text{strips of width } \Delta x_i$$



$$\sum_k \Delta y_k m\{x : f(x) > y_k\} \quad \text{bands of height } \Delta y_k$$

For the nonnegative function shown, both constructions give the same integral.

The Lebesgue integral is defined for a larger class of functions; these slides use the Riemann picture. Taking m to be a probability turns the band sum into an expectation.

Video: [vcubingx, A horizontal integral? Introduction to Lebesgue Integration.](#)

The area function

Let f be **continuous** on an interval containing a and x , and define

$$A(x) = \int_a^x f(t) dt.$$

FTC I. $A'(x) = f(x)$.

If $f \geq 0$, $A(x)$ is the accumulated area under the graph.

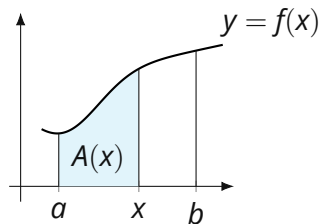


Figure 1 · The area function $A(x)$.

$A(x)$ accumulates signed area up to x .

Why $A'(x) = f(x)$

For $h > 0$, let $m_h = \min_{t \in [x, x+h]} f(t)$ and $M_h = \max_{t \in [x, x+h]} f(t)$. Then

$$m_h h \leq A(x+h) - A(x) \leq M_h h.$$

Dividing by h gives

$$m_h \leq \frac{A(x+h) - A(x)}{h} \leq M_h.$$

By continuity, $m_h, M_h \rightarrow f(x)$ as $h \rightarrow 0^+$. The same argument with the interval $[x+h, x]$ handles $h < 0$. Hence $A'(x) = f(x)$.

Video: [Dr. Trefor Bazett, Fundamental theorem of calculus 1: the geometric idea.](#)

FTC I

Differentiate each accumulation function.

1. $A(x) = \int_a^x (t^2 + 1) dt.$

2. $B(x) = \int_1^{x^2} \ln(1 + t) dt.$

3. $C(x) = \int_x^2 e^{-t^2} dt.$

4. $D(x) = \int_{\sin x}^x e^{t^2} dt.$

FTC I

Apply FTC I and the chain rule when an endpoint is a function of x .

$$A'(x) = x^2 + 1,$$

$$B'(x) = \ln(1 + x^2) \cdot 2x,$$

$$C'(x) = -e^{-x^2},$$

$$D'(x) = e^{x^2} - e^{\sin^2 x} \cos x.$$

Video: 3Blue1Brown, *Integration and the fundamental theorem of calculus*.

Antiderivatives

To compute a definite integral, find an antiderivative and evaluate it at the end-points.

Antiderivative. On an interval, F is an antiderivative of f if $F' = f$. The indefinite integral is the family

$$\int f(x) dx = F(x) + C.$$

Common antiderivatives

Each formula holds on every interval where its integrand is defined:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1,$$

$$\int \frac{1}{x} dx = \ln |x| + C,$$

$$\int e^x dx = e^x + C,$$

$$\int a^x dx = \frac{a^x}{\ln a} + C, \quad a > 0, a \neq 1.$$

$$\int \cos x dx = \sin x + C,$$

$$\int \sin x dx = -\cos x + C,$$

$$\int \sec^2 x dx = \tan x + C,$$

$$\int \frac{1}{1+x^2} dx = \arctan x + C.$$

Differentiate each right-hand side to check the corresponding integrand.

Video: [Professor Dave Explains, *Evaluating indefinite integrals*](#).

Common antiderivatives

Find one antiderivative of each function.

1. $6x^5 - 4x + 3$.

2. $x^{-3} + 2x^{-1}, x \neq 0$.

3. $3e^x - 2\cos x$.

4. $5\sin x + 4\sec^2 x$.

5. $2^x + \frac{1}{1+x^2}$.

6. $\frac{3}{\sqrt{x}}, x > 0$.

Common antiderivatives

$$\int (6x^5 - 4x + 3) dx = x^6 - 2x^2 + 3x + C,$$

$$\int (x^{-3} + 2x^{-1}) dx = -\frac{1}{2x^2} + 2 \ln |x| + C,$$

$$\int (3e^x - 2 \cos x) dx = 3e^x - 2 \sin x + C.$$

$$\int (5 \sin x + 4 \sec^2 x) dx = -5 \cos x + 4 \tan x + C,$$

$$\int \left(2^x + \frac{1}{1+x^2} \right) dx = \frac{2^x}{\ln 2} + \arctan x + C,$$

$$\int \frac{3}{\sqrt{x}} dx = 6\sqrt{x} + C.$$

From primitives to the definite integral

Suppose f is **continuous** on $[a, b]$ and $F' = f$ on an interval containing $[a, b]$. If $A(x) = \int_a^x f(t) dt$, FTC I gives $A'(x) = f(x) = F'(x)$, so $A - F$ is constant. Since $A(a) = 0$,

FTC II.

$$\int_a^b f(x) dx = F(b) - F(a).$$

$$f \xrightarrow{\text{accumulate}} A(x) = \int_a^x f(t) dt \xrightarrow{\text{differentiate}} f(x)$$

FTC I says differentiation recovers the current rate from accumulated change. FTC II says that if $F' = f$, accumulated change from a to b is $F(b) - F(a)$. Thus a **definite integral** is an accumulated quantity, whereas an **indefinite integral** denotes **the family of primitives** $F + C$.

For an absolutely continuous F , F' exists almost everywhere and $F(b) - F(a) = \int_a^b F'(x) dx$.

FTC II

Evaluate each definite integral using an antiderivative.

1. $\int_0^2 (3x^2 - 4x + 1) dx.$

2. $\int_1^e \frac{1}{x} dx.$

3. $\int_0^{\pi/2} \cos x dx.$

4. $\int_2^4 q^3 dq.$

FTC II

$$\int_0^2 (3x^2 - 4x + 1) dx = [x^3 - 2x^2 + x]_0^2 = 2,$$

$$\int_1^e \frac{1}{x} dx = [\ln x]_1^e = 1,$$

$$\int_0^{\pi/2} \cos x dx = [\sin x]_0^{\pi/2} = 1,$$

$$\int_2^4 q^3 dq = \left[\frac{q^4}{4} \right]_2^4 = 60.$$

Video: [MIT 18.01SC, Second fundamental theorem and chain rule.](#)

A marginal-cost calculation

Accumulating marginal cost

Example. If $MC(q) = q^3$, then the variable cost of raising output from 2 to 4 is

$$\int_2^4 MC(q) dq = \int_2^4 q^3 dq = \left[\frac{q^4}{4} \right]_2^4 = 60.$$

Integration accumulates marginal cost over the output interval.

Signed vs. geometric area

Where $f < 0$, the definite integral counts *negative* signed area. The geometric area uses $\int_a^b |f(x)| dx$, splitting at the roots:

$$\int_a^{p_1} -f + \int_{p_1}^{p_2} f + \int_{p_2}^{p_3} -f + \int_{p_3}^b f.$$

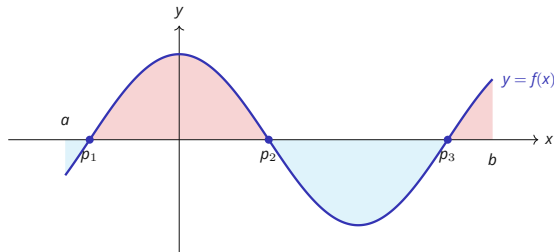


Figure 2 · Signed area.

The sign of the integrand changes at the roots

p_1, p_2, p_3 .

Signed and geometric area

Let $f(x) = x^2 - 1$ on $[-2, 2]$.

(a) Find the roots and compute $\int_{-2}^2 f(x) dx$.

(b) Split the interval at the roots and compute the geometric area $\int_{-2}^2 |f(x)| dx$.

Signed and geometric area

The roots are $x = \pm 1$. Evenness gives

$$\int_{-2}^2 (x^2 - 1) dx = 2 \left[\frac{x^3}{3} - x \right]_0^2 = \frac{4}{3}.$$

The sign changes at ± 1 , so

$$\int_{-2}^2 |x^2 - 1| dx = 2 \left(\int_0^1 (1 - x^2) dx + \int_1^2 (x^2 - 1) dx \right) = 2 \left(\frac{2}{3} + \frac{4}{3} \right) = 4.$$

Section 2

Properties of integrals

Indefinite integrals

For $F' = f$ and $G' = g$ on an interval,

$$\int f(x) dx = F(x) + C.$$

◆ **Linearity:** $\int (af + bg) dx = a \int f dx + b \int g dx.$

◆ Constants of integration are absorbed into a single arbitrary constant.

Definite integrals (i)

For continuous f, g on an interval and compatible endpoints:

1. $\int_a^a f dx = 0.$
2. $\int_a^b f dx = - \int_b^a f dx.$
3. $\int_a^b (\alpha f + \beta g) dx = \alpha \int_a^b f dx + \beta \int_a^b g dx.$
4. $\int_a^b f dx = \int_a^c f dx + \int_c^b f dx.$ Split at points where the formula or the sign of the integrand changes.
5. $\frac{d}{dx} \int_a^x f(z) dz = f(x)$ and $\frac{d}{dx} \int_x^b f(z) dz = -f(x).$

Video: [MAT137, Five properties of definite integrals.](#)

Definite integrals (ii)

For continuous functions on $[a, b]$ with $b > a$:

6. If $f(x) \geq g(x)$ on $[a, b]$, then $\int_a^b f dx \geq \int_a^b g dx$.

7. $\left| \int_a^b f dx \right| \leq \int_a^b |f| dx$.

8. **Cauchy-Schwarz:**

$$\left| \int_a^b f(x)g(x) dx \right|^2 \leq \left(\int_a^b f(x)^2 dx \right) \left(\int_a^b g(x)^2 dx \right).$$

Definite integral properties

Suppose

$$\int_a^b f(x) dx = 3, \quad \int_b^c f(x) dx = -1, \quad \int_a^c g(x) dx = 4.$$

Compute:

1. $\int_a^c f(x) dx$
2. $\int_c^a f(x) dx$
3. $\int_a^c (2f(x) - g(x)) dx$

If additionally $\int_a^c f(x)^2 dx = 4$ and $\int_a^c g(x)^2 dx = 9$, give the Cauchy-Schwarz bound for $\left| \int_a^c f(x)g(x) dx \right|$.

Worked calculations

$$\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx = 3 + (-1) = 2,$$

$$\int_c^a f(x) dx = - \int_a^c f(x) dx = -2,$$

$$\int_a^c (2f(x) - g(x)) dx = 2 \int_a^c f(x) dx - \int_a^c g(x) dx = 2(2) - 4 = 0.$$

By Cauchy-Schwarz,

$$\left| \int_a^c f(x)g(x) dx \right| \leq \left(\int_a^c f(x)^2 dx \right)^{1/2} \left(\int_a^c g(x)^2 dx \right)^{1/2} = \sqrt{4} \sqrt{9} = 6.$$

Average value of a function

Average value and equal accumulation

For a continuous function f on $[a, b]$ with $b > a$, define

Average value.

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx.$$

The constant function f_{avg} has the same total accumulation:

$$(b-a)f_{\text{avg}} = \int_a^b f(x) dx.$$

Video: Professor Dave Explains, *The mean value theorem for integrals and average value.*

Average value of a function

Integral mean-value result

If $m = \min_{[a,b]} f$ and $M = \max_{[a,b]} f$, order preservation gives

$$m \leq f_{\text{avg}} \leq M.$$

By continuity and the Intermediate Value Theorem, there is some $c \in [a, b]$ such that

$$f(c) = f_{\text{avg}}.$$

This reconnects integration to the continuity results from Functions: continuity gives extrema on $[a, b]$ and ensures that every intermediate value, including the average value, is attained.

Further reading: [OpenStax, Calculus Volume 1, Section 5.2](#).

The average value is attained

Attaining the average value

For

$$f(x) = (x - 2)^3 - 3(x - 2) + 3$$

on $[0, 4]$,

$$\int_0^4 f(x) dx = 12, \quad f_{\text{avg}} = 3,$$

with $m = 1$ and $M = 5$. Solving

$$f(x) = 3 \text{ gives } x = 2 \text{ and } x = 2 \pm \sqrt{3}.$$

The shaded region and the dashed rectangle of height f_{avg} both have area 12. The theorem supplies one such c ; here the level is met three times.

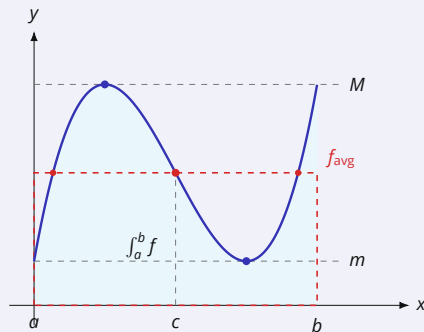


Figure 3 · The level f_{avg} lies between m and M ; the graph meets it at c and at the two smaller dots.

Average value and where it is attained

For $f(x) = x^2$ on $[0, 2]$, compute f_{avg} and find every $c \in [0, 2]$ satisfying $f(c) = f_{\text{avg}}$.

Average value and where it is attained

$$f_{\text{avg}} = \frac{1}{2-0} \int_0^2 x^2 dx = \frac{1}{2} \left[\frac{x^3}{3} \right]_0^2 = \frac{4}{3}.$$

Solving $c^2 = 4/3$ and imposing $c \in [0, 2]$ gives

$$c = \frac{2}{\sqrt{3}}.$$

Using additivity and orientation

Suppose

$$\int_0^1 f(x) dx = 2, \quad \int_1^3 f(x) dx = -1.$$

Additivity gives

$$\int_0^3 f(x) dx = \int_0^1 f(x) dx + \int_1^3 f(x) dx = 2 + (-1) = 1.$$

Reversing the endpoints changes the sign:

$$\int_3^0 f(x) dx = - \int_0^3 f(x) dx = -1.$$

These conclusions follow from the order and linearity properties of the integral, without computing an antiderivative.

Section 3

Differentiation of integrals

Where the parameter enters

Consider the parameterized accumulation

$$G(y) = \int_{a(y)}^{b(y)} f(x, y) dx.$$

As y changes, $G(y)$ can change for three distinct reasons:

1. the **integrand** changes through $f(x, y)$;
2. the **upper bound** moves through $b(y)$;
3. the **lower bound** moves through $a(y)$.

Two earlier-looking formulas are special cases:

$$\frac{d}{dy} \int_a^b f(x, y) dx = \int_a^b f_y(x, y) dx \quad (\text{fixed bounds}),$$

$$\frac{d}{dy} \int_a^y f(x) dx = f(y) \quad (\text{FTC I}).$$

The next formula combines the interior effect and both boundary effects.

Leibniz's rule

Suppose $a(y), b(y)$ are differentiable and the needed partial derivatives are continuous. Then

Leibniz rule.

$$\frac{d}{dy} \int_{a(y)}^{b(y)} f(x, y) dx = \int_{a(y)}^{b(y)} \frac{\partial f}{\partial y}(x, y) dx + f(b(y), y)b'(y) - f(a(y), y)a'(y).$$

Moving the bounds creates the two boundary terms; changing the integrand creates the integral term. FTC I and differentiation under fixed limits are therefore special cases of the same rule.

Video: [Brian Storey, Leibniz integral rule.](#)

A compact check

Example. For

$$G(y) = \int_0^{y^2} e^{-xy} dx,$$

the lower endpoint is fixed, the integrand depends on y , and the upper endpoint is y^2 .
Leibniz's rule gives

$$G'(y) = \int_0^{y^2} (-x)e^{-xy} dx + e^{-y^3} \cdot 2y.$$

The first term is the interior effect; the second is the moving-boundary effect.

Leibniz's rule

Differentiate each parameterized integral.

1. $G(y) = \int_0^1 (x^2 + y^2) dx.$

2. $H(y) = \int_0^{y^2} \cos x dx.$

3. $K(y) = \int_y^{2y} (x + y) dx.$

4. $L(y) = \int_{\sin y}^{y^2} e^{x+y} dx.$

Leibniz's rule: items 1 and 2

Take f , a and b from the integral, then apply:

$$\frac{d}{dy} \int_{a(y)}^{b(y)} f(x, y) dx = \int_{a(y)}^{b(y)} f_y(x, y) dx + f(b(y), y) b'(y) - f(a(y), y) a'(y).$$

1. $G(y) = \int_0^1 (x^2 + y^2) dx$

$a = 0$ and $b = 1$ are constant, so $a' = b' = 0$ and only the integral term survives:

$$\begin{aligned} G'(y) &= \int_0^1 \frac{\partial}{\partial y} (x^2 + y^2) dx \\ &= \int_0^1 2y dx = 2y [x]_0^1 = 2y. \end{aligned}$$

2. $H(y) = \int_0^{y^2} \cos x dx$

The integrand $\cos x$ does not contain y , so $f_y = 0$ and only the upper limit $b(y) = y^2$ moves:

$$\begin{aligned} H'(y) &= \underbrace{\int_0^{y^2} 0 dx}_{=0} + \cos(y^2) \cdot \frac{d}{dy} y^2 \\ &= 2y \cos(y^2). \end{aligned}$$

Item 1 is the integral term alone, item 2 the boundary term alone.

Leibniz's rule: items 3 and 4

3. $K(y) = \int_y^{2y} (x+y) dx$

Here $f_y = 1$, $a(y) = y$ with $a' = 1$, and $b(y) = 2y$ with $b' = 2$. The boundary values are $f(2y, y) = 3y$ and $f(y, y) = 2y$:

$$\begin{aligned} K'(y) &= \int_y^{2y} 1 dx + (3y)(2) - (2y)(1) \\ &= (2y - y) + 6y - 2y = 5y. \end{aligned}$$

Check: $K(y) = \left[\frac{x^2}{2} + xy \right]_y^{2y} = \frac{5}{2}y^2$.

4. $L(y) = \int_{\sin y}^{y^2} e^{x+y} dx$

Now $f_y = e^{x+y}$, $b(y) = y^2$ with $b' = 2y$, and $a(y) = \sin y$ with $a' = \cos y$:

$$\begin{aligned} L'(y) &= \int_{\sin y}^{y^2} e^{x+y} dx \\ &\quad + e^{y^2+y} (2y) - e^{\sin y+y} (\cos y). \end{aligned}$$

The remaining integral factors as

$$\int_{\sin y}^{y^2} e^{x+y} dx = e^y [e^x]_{\sin y}^{y^2} = e^{y^2+y} - e^{\sin y+y}.$$

Video: [Dr. Chris Tisdell, Leibniz rule for differentiating under an integral sign.](#)

Price changes market share through two margins

A market-share derivative

Consumers have type density $g(t)$ on $[0, 1]$. A type $t \leq q(p)$ buys $d(t, p)$ units, where $q(p) \in [0, 1]$ and $q'(p) < 0$. Aggregate demand is

$$D(p) = \int_0^{q(p)} d(t, p) g(t) dt.$$

Leibniz's rule gives

$$D'(p) = \int_0^{q(p)} d_p(t, p) g(t) dt + d(q(p), p) g(q(p)) q'(p).$$

The integral measures the response of inframarginal consumers; the boundary term counts consumers who enter or leave as the cutoff moves.

A distributional parameter changes expected utility

Expected utility with a parameter

Let income have density $f(y; \theta)$ on the fixed support $[0, \bar{y}]$. Expected utility is

$$EU(\theta) = \int_0^{\bar{y}} u(y) f(y; \theta) dy.$$

Under the continuity assumptions stated in Leibniz's rule,

$$EU'(\theta) = \int_0^{\bar{y}} u(y) \frac{\partial f(y; \theta)}{\partial \theta} dy.$$

The derivative weights the density change at each income by the utility attached to that income.

Eligibility changes welfare at the intensive and extensive margins

Welfare with a moving eligibility cutoff

A transfer covers incomes $y \leq c(\tau)$, where $c'(\tau) > 0$. With welfare weight $w(y, \tau)$ and income density $f(y)$,

$$W(\tau) = \int_0^{c(\tau)} w(y, \tau) f(y) dy.$$

Hence

$$W'(\tau) = \int_0^{c(\tau)} w_\tau(y, \tau) f(y) dy + w(c(\tau), \tau) f(c(\tau)) c'(\tau).$$

The integral changes welfare for agents already eligible; the boundary term adds the agents who cross the moving cutoff.

A posted price trades off revenue and sales probability

Posted-price revenue

A buyer has valuation V with density f on $[0, \bar{v}]$ and buys one unit when $V \geq r$. Expected revenue from price r is

$$R(r) = \int_r^{\bar{v}} rf(v) dv = r[1 - F(r)].$$

Differentiating the integrand and lower bound gives

$$R'(r) = \int_r^{\bar{v}} f(v) dv - rf(r) = 1 - F(r) - rf(r).$$

The first term is the gain when a sale occurs; the second is the revenue lost when marginal buyers are excluded.

Section 4

Integration rules

Reverse the derivative rules

The two rules below reverse familiar differentiation rules.

Chain rule $\leftrightarrow$ substitution.

$$\frac{d}{dx}\Phi(g(x)) = \Phi'(g(x))g'(x)$$

becomes

$$\int \Phi'(g(x))g'(x) dx = \Phi(g(x)) + C.$$

Product rule $\leftrightarrow$ integration by parts.

$$(uv)' = u'v + uv'$$

becomes

$$\int uv' = uv - \int u'v.$$

Immediate integrals (i)

Each rule reverses a chain-rule derivative: spot the inner function f and its factor f' .

Power and exponential.

$$\int f'(x) f(x)^n dx = \frac{f(x)^{n+1}}{n+1} + C, \quad n \neq -1$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + C$$

Logarithm and general base.

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int a^{f(x)} f'(x) dx = \frac{a^{f(x)}}{\ln a} + C$$

Conditions: the power formula applies on intervals where $f(x)^n$ is real and differentiable; the logarithm formula needs $f(x) \neq 0$; the general-base formula needs $a > 0$ and $a \neq 1$.

Immediate integrals (ii)

$$\int \sin(f(x))f'(x) dx = -\cos(f(x)) + C$$

$$\int \cos(f(x))f'(x) dx = \sin(f(x)) + C$$

$$\int \ln |f(x)|f'(x) dx = f(x)(\ln |f(x)| - 1) + C, \quad f(x) \neq 0$$

Recognising the inner derivative

Each integrand contains a familiar outer function together with the derivative of its inner argument:

$$\int \frac{2x}{1+x^2} dx = \ln(1+x^2) + C,$$

$$\int 3e^{3x-1} dx = e^{3x-1} + C,$$

$$\int 2 \cos(2x) dx = \sin(2x) + C.$$

A reliable check is to differentiate the proposed antiderivative and recover the original integrand.

Substitution

Substitution is the integral version of the chain rule.

Change of variable. If $u = g(x)$ and $du = g'(x)dx$, then

$$\int \varphi(g(x))g'(x) dx = \int \varphi(u) du.$$

Example.

$$d(x^2) = 2x dx,$$

$$u = x^2,$$

$$du = d(x^2) = 2x dx,$$

$$\int 2xe^{x^2} dx = \int e^u du = e^u + C = e^{x^2} + C.$$

Video: [Dr. Trefor Bazett, Intro to Substitution: Undoing the Chain Rule.](#)

Substitution with definite limits

If $u = g(x)$, then the endpoints must be transformed together with the variable:

Definite substitution.

$$\int_a^b \varphi(g(x))g'(x) dx = \int_{g(a)}^{g(b)} \varphi(u) du.$$

Example. With $u = x^2$, the bounds $x = 0, 1$ become $u = 0, 1$:

$$\int_0^1 2xe^{x^2} dx = \int_0^1 e^u du = [e^u]_0^1 = e - 1.$$

Either change the bounds immediately, as above, or return to the original variable before evaluating the original endpoints.

Video: [Radford Mathematics, U-substitution for definite integrals.](#)

Integration by parts

With $u = u(x)$ and $dv = v'(x)dx$,

$$\int_a^b u(x)v'(x) dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) dx.$$

It comes from the product rule.

Example. Taking $u = x$ and $dv = e^{6x}dx$,

$$\int xe^{6x} dx = \frac{x}{6}e^{6x} - \frac{1}{6} \int e^{6x} dx = \frac{x}{6}e^{6x} - \frac{1}{36}e^{6x} + C.$$

Video: [Dr. Trefor Bazett, Integration by Parts.](#)

Selecting an antiderivative method

Selecting an integration method

Evaluate each integral and identify the method before calculating:

(a) $\int \sqrt{x} \, dx$

(c) $\int \frac{1}{\sqrt{4x-1}} \, dx$ on $x > \frac{1}{4}$

(b) $\int x e^{2x} \, dx$

(d) $\int x \cos(x^2) \, dx$

The four integrals require a power antiderivative, integration by parts, and substitution.

Four antiderivatives

Four antiderivative calculations

POWER RULE

$$\int \sqrt{x} \, dx = \int x^{1/2} \, dx = \frac{2}{3} x^{3/2} + C.$$

INTEGRATION BY PARTS

With $u = x$ and $dv = e^{2x} dx$,

$$\int x e^{2x} dx = \frac{x e^{2x}}{2} - \frac{e^{2x}}{4} + C.$$

SUBSTITUTION

With $u = 4x - 1$,

$$\int \frac{dx}{\sqrt{4x-1}} = \frac{1}{2} \sqrt{4x-1} + C, \quad x > \frac{1}{4}.$$

SUBSTITUTION

With $u = x^2$,

$$\int x \cos(x^2) \, dx = \frac{1}{2} \sin(x^2) + C.$$

Further exercises: [MIT 18.01SC, Session 85](#).

More integration by parts

Evaluate, showing the choice of u and dv :

$$(a) \int x \ln x \, dx \quad (x > 0), \quad (b) \int x^2 e^x \, dx.$$

More integration by parts

For (a), choose $u = \ln x$ and $dv = x dx$:

$$\int x \ln x dx = \frac{x^2}{2} \ln x - \frac{1}{2} \int x dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C.$$

For (b), apply integration by parts twice:

$$\int x^2 e^x dx = x^2 e^x - 2 \int x e^x dx = e^x (x^2 - 2x + 2) + C.$$

Section 5

Improper integrals

Ordinary and improper integrals

Ordinary definite integral. A continuous function on a closed interval is Riemann integrable. Bounded functions with finitely many jump discontinuities are also Riemann integrable.

Improper integral. Infinite endpoints and singularities are handled by limits of ordinary definite integrals. An interior singularity requires convergence from both sides.

Further reading: [OpenStax, Calculus Volume 2, Section 3.7](#).

Defined as a limit

An improper integral is built from ordinary definite integrals by taking a limit:

$$\int_a^\infty f(x) dx := \lim_{b \rightarrow \infty} \int_a^b f(x) dx.$$

If $F' = f$ on every finite truncation, FTC II gives

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} (F(b) - F(a)),$$

provided this limit exists and is finite.

- ◆ $\int_1^\infty \frac{1}{x} dx$ diverges.
- ◆ $\int_1^\infty \frac{1}{x^2} dx = 1.$

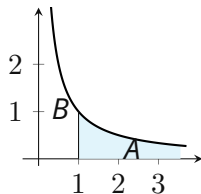


Figure 4 · An expanding integration interval.

First integrate over a finite interval; then let the endpoint tend to infinity.

Definition

An improper integral has an infinite interval or an integrand that is unbounded or undefined at an endpoint or interior point. If $c \in (a, b)$ is an interior singularity, define

$$\int_a^b f = \lim_{z \rightarrow c^-} \int_a^z f + \lim_{z \rightarrow c^+} \int_z^b f,$$

and require both one-sided integrals to converge. At an endpoint or at $\pm\infty$, a single one-sided limit defines the value.

Endpoint singularities and tails

Typical behaviour near a singularity or far out in the tail:

- ◆ $\int_0^1 \frac{1}{x} dx = \lim_{z \rightarrow 0^+} \int_z^1 \frac{1}{x} dx = \infty.$
- ◆ $\int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{z \rightarrow 0^+} \int_z^1 x^{-1/2} dx = 2.$
- ◆ $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$ (Gaussian integral).

The Gaussian integral exists although its integrand has no elementary antiderivative. Its value is determined by the defining limit.

Power-integral convergence test

For power tails,

Tail test.

$$\int_1^{\infty} \frac{1}{x^p} dx = \begin{cases} \frac{1}{p-1}, & p > 1, \\ +\infty, & p \leq 1. \end{cases}$$

Near-zero test.

$$\int_0^1 \frac{1}{x^p} dx = \begin{cases} \frac{1}{1-p}, & p < 1, \\ +\infty, & p \geq 1. \end{cases}$$

Convergence of improper integrals

Decide whether each integral converges and compute its value when it does.

1. $\int_1^{\infty} x^{-2} dx.$

2. $\int_1^{\infty} x^{-1} dx.$

3. $\int_0^1 x^{-1/2} dx.$

4. $\int_0^1 x^{-1} dx.$

Video: [MIT 18.01SC, *Improper Integrals*](#). Exercises: [MIT 18.01SC, Session 91](#).

Convergence of improper integrals

1.

$$\begin{aligned}\int_1^{\infty} x^{-2} dx &= \lim_{R \rightarrow \infty} \int_1^R x^{-2} dx \\ &= \lim_{R \rightarrow \infty} \left[-\frac{1}{x} \right]_1^R = \lim_{R \rightarrow \infty} \left(1 - \frac{1}{R} \right) = 1.\end{aligned}$$

2.

$$\begin{aligned}\int_1^{\infty} x^{-1} dx &= \lim_{R \rightarrow \infty} \int_1^R \frac{dx}{x} \\ &= \lim_{R \rightarrow \infty} [\ln x]_1^R = \lim_{R \rightarrow \infty} \ln R = +\infty.\end{aligned}$$

Hence item 2 diverges.

3.

$$\begin{aligned}\int_0^1 x^{-1/2} dx &= \lim_{\varepsilon \rightarrow 0^+} \int_{\varepsilon}^1 x^{-1/2} dx \\ &= \lim_{\varepsilon \rightarrow 0^+} [2\sqrt{x}]_{\varepsilon}^1 = \lim_{\varepsilon \rightarrow 0^+} (2 - 2\sqrt{\varepsilon}) = 2.\end{aligned}$$

4.

$$\begin{aligned}\int_0^1 x^{-1} dx &= \lim_{\varepsilon \rightarrow 0^+} \int_{\varepsilon}^1 \frac{dx}{x} \\ &= \lim_{\varepsilon \rightarrow 0^+} [\ln x]_{\varepsilon}^1 = \lim_{\varepsilon \rightarrow 0^+} (-\ln \varepsilon) = +\infty.\end{aligned}$$

Hence item 4 diverges.

Check before applying FTC II

An interior singularity

A tempting calculation is

$$\int_0^2 \frac{1}{(x-1)^2} dx \stackrel{?}{=} \left[-\frac{1}{x-1} \right]_0^2 = -2.$$

1. What is wrong with this calculation?
2. Where must the integral be split?
3. Do the two one-sided improper integrals converge?

The singularity cannot be crossed

Check the domain before using FTC II

The integrand is undefined at $x = 1$, so this is not an ordinary definite integral on $[0, 2]$. It must be split at the singularity:

LEFT SIDE

$$\lim_{a \rightarrow 1^-} \int_0^a \frac{dx}{(x-1)^2} = \lim_{a \rightarrow 1^-} \left(-\frac{1}{a-1} - 1 \right) = +\infty.$$

RIGHT SIDE

$$\lim_{b \rightarrow 1^+} \int_b^2 \frac{dx}{(x-1)^2} = \lim_{b \rightarrow 1^+} \left(-1 + \frac{1}{b-1} \right) = +\infty.$$

Both one-sided integrals diverge. The primitive $-1/(x-1)$ is valid on each side of 1, but it cannot be evaluated across the singularity as though the integrand were continuous on the whole interval.

$$\int_0^2 \frac{dx}{(x-1)^2} = +\infty.$$

Thus the improper integral diverges and has no finite value.

Section 6

Several variables

Multiple integrals over rectangles

A one-dimensional integral accumulates local contributions “height $\times$ width.” A multiple integral uses the same idea over area or volume. For

$$A = \prod_{i=1}^n [a_i, b_i], \quad f: A \rightarrow \mathbb{R}$$

continuous, Fubini lets us compute that single multiple integral by repeated one-dimensional integrations:

$$\int_A f(x) dx = \int_{a_n}^{b_n} \cdots \int_{a_1}^{b_1} f(x_1, \dots, x_n) dx_1 \cdots dx_n.$$

Video: [DrOfEng, Double integrals over rectangular regions and Fubini's theorem.](#)

Inner and outer integration

For a rectangle $R = [a, b] \times [c, d]$,

$$\int_a^b \int_c^d f(x, y) dy dx$$

is evaluated from the inside out.

- ◆ **Inner integral:** fix $x = \bar{x}$ and accumulate along the vertical slice $c \leq y \leq d$.
- ◆ **Outer integral:** let the slice move from $x = a$ to $x = b$ and accumulate the resulting slice values.
- ◆ **Order:** for **continuous** f on the rectangle, Fubini allows the order of integration to be changed.
- ◆ **Separability:** if $f(x, y) = g(x)h(y)$, the double integral factors into the product of two one-dimensional integrals.

Video: [Dr. Trefor Bazett, Changing the order of integration.](#)

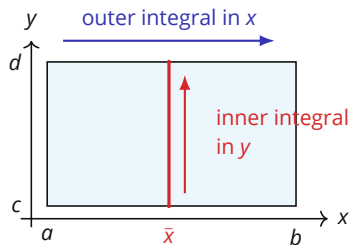


Figure 5 · Inner integration runs along one slice before the outer integral moves across the rectangle.

A Fubini calculation

Changing the order on a rectangle

Let $R = [0, 1] \times [0, 2]$ and $f(x, y) = x + 2y$.

INTEGRATE y FIRST

$$\begin{aligned} \int_0^1 \int_0^2 (x + 2y) \, dy \, dx &= \int_0^1 [xy + y^2]_0^2 \, dx \\ &= \int_0^1 (2x + 4) \, dx \\ &= 5. \end{aligned}$$

INTEGRATE x FIRST

$$\begin{aligned} \int_0^2 \int_0^1 (x + 2y) \, dx \, dy &= \int_0^2 \left(\frac{1}{2} + 2y \right) \, dy \\ &= \left[\frac{y}{2} + y^2 \right]_0^2 \\ &= 5. \end{aligned}$$

Continuity on the rectangle justifies either order; both iterated integrals compute the same double integral.

Accumulation over rectangles

Two iterated integrals

Evaluate each iterated integral in the written order:

$$I_1 = \int_0^1 \int_1^e \left(2x + \frac{1}{y} \right) dy dx,$$

$$I_2 = \int_1^2 \int_0^3 (xy + \ln x) dy dx.$$

At the inner step, find an antiderivative and substitute both limits before moving to the outer integral.

Accumulation over rectangles

Two iterated integrals

$$\begin{aligned} I_1 &= \int_0^1 [2xy + \ln y]_1^e dx \\ &= \int_0^1 (2x(e-1) + 1) dx \\ &= e. \end{aligned}$$

$$\begin{aligned} I_2 &= \int_1^2 \left[\frac{xy^2}{2} + y \ln x \right]_0^3 dx \\ &= \int_1^2 \left(\frac{9}{2}x + 3 \ln x \right) dx \\ &= \frac{15}{4} + 6 \ln 2. \end{aligned}$$

FTC in several variables

This is FTC I applied one coordinate at a time. If f is continuous, repeated integration and repeated differentiation undo each other:

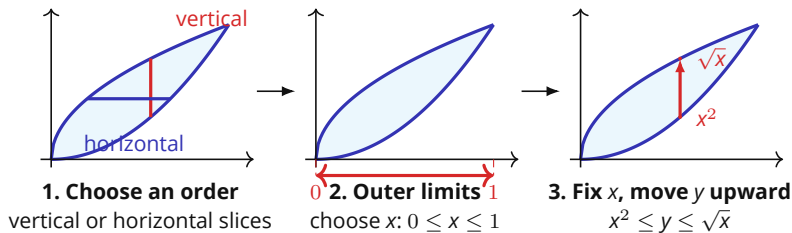
$$\frac{\partial^n}{\partial x_1 \cdots \partial x_n} \int_{a_n}^{x_n} \cdots \int_{a_1}^{x_1} f(t_1, \dots, t_n) dt_1 \cdots dt_n = f(x_1, \dots, x_n).$$

Regularity

If f is continuous on $[a, b] \times [c, d]$, the two iterated integrals both equal $\iint f dA$. For $F(\theta) = \int_a^b f(x, \theta) dx$, if f and f_θ are continuous for $x \in [a, b]$ and $\theta \in [\theta_0, \theta_1]$, then $F'(\theta) = \int_a^b f_\theta(x, \theta) dx$.

Finding limits from a region

Unless an order is specified, choose it. Prefer slices that meet the region in one connected piece and give simpler limits.



x outer $\Rightarrow$ vertical slice $\Rightarrow dy$ first

y outer $\Rightarrow$ horizontal slice $\Rightarrow dx$ first

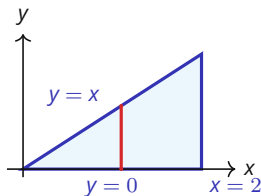
If a slice meets the region in more than one piece, split the region or choose the other order.

Setting limits in both directions

Each region is given by its boundary curves. The chosen outer variable determines the slice direction.

CHOOSE x OUTER

R is bounded by $y = 0$, $y = x$, and $x = 2$.

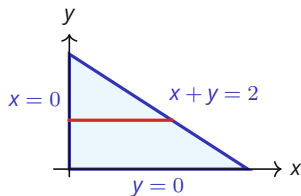


$$0 \leq x \leq 2, \quad 0 \leq y \leq x.$$

The inner variable runs along the slice from its first boundary point to its second.

CHOOSE y OUTER

R is bounded by $x = 0$, $y = 0$, and $x + y = 2$.



$$0 \leq y \leq 2, \quad 0 \leq x \leq 2 - y.$$

Non-rectangular domains: vertical slices

Fix x , then move upward along the vertical slice.

$$A = \{(x, y) : a \leq x \leq b, h(x) \leq y \leq g(x)\}.$$

The inner limits are $y = h(x)$ and $y = g(x)$:

$$\int_A f(x, y) dA = \int_a^b \int_{h(x)}^{g(x)} f(x, y) dy dx.$$

For the pictured region,

$$0 \leq x \leq 1, \quad x^2 \leq y \leq \sqrt{x}.$$

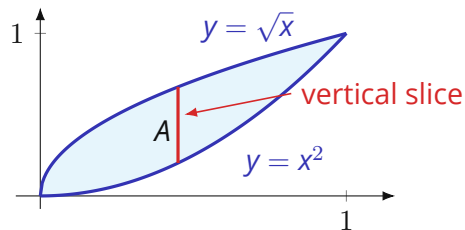


Figure 6 · The lower and upper endpoints give the inner limits.

Vertical slices: a worked integral

The curves meet at $x = 0$ and $x = 1$. At each fixed x , move from $y = x^2$ to $y = \sqrt{x}$.

$$\int_0^1 \int_{x^2}^{\sqrt{x}} y \, dy \, dx = \int_0^1 \left[\frac{y^2}{2} \right]_{x^2}^{\sqrt{x}} dx = \frac{1}{2} \int_0^1 (x - x^4) \, dx = \frac{3}{20}.$$

Write both orders of integration

- (a) For $R = \{(x, y) : 0 \leq y \leq x, x \leq 2\}$, write $\iint_R f(x, y) dA$ in the orders $dy dx$ and $dx dy$.
- (b) Let D be the region between $y = x^2$ and $y = 2x$. Find the intersection points and write both orders of integration.

Write both orders of integration

For the triangle R ,

$$\iint_R f dA = \int_0^2 \int_0^x f(x,y) dy dx = \int_0^2 \int_y^2 f(x,y) dx dy.$$

For D , the curves meet where $x^2 = 2x$, hence at $(0,0)$ and $(2,4)$. Therefore

$$\iint_D f dA = \int_0^2 \int_{x^2}^{2x} f(x,y) dy dx = \int_0^4 \int_{y/2}^{\sqrt{y}} f(x,y) dx dy.$$

Change of variables and the Jacobian

Area rescaling under a coordinate map

Suppose $T(u, v) = (x(u, v), y(u, v))$ maps B one-to-one onto A , is continuously differentiable, and has nonzero Jacobian determinant in the interior. With

$$J_T(u, v) = \det \begin{pmatrix} \partial x / \partial u & \partial x / \partial v \\ \partial y / \partial u & \partial y / \partial v \end{pmatrix},$$

$$\int_A f(x, y) dA = \int_B f(x(u, v), y(u, v)) |J_T(u, v)| du dv.$$

Further reading: [OpenStax, Calculus Volume 3, Section 5.7](#). Video: [Dr. Trefor Bazett, Change of Variables and the Jacobian](#).

Polar coordinates

Jacobian of the polar map

For $x = r \cos \theta$ and $y = r \sin \theta$,

$$J_T(r, \theta) = \det \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} = r.$$

Hence $dA = r dr d\theta$.

A complete polar integral

Integrating $x^2 + y^2$ over the unit disk

On $x^2 + y^2 \leq 1$, use $x^2 + y^2 = r^2$ and $dA = r dr d\theta$:

$$\iint_{x^2+y^2 \leq 1} (x^2 + y^2) dA = \int_0^{2\pi} \int_0^1 r^2 r dr d\theta = 2\pi \left[\frac{r^4}{4} \right]_0^1 = \frac{\pi}{2}.$$

The factor r^3 combines the transformed integrand r^2 with the Jacobian factor r .

Polar area and linear rescaling

- (a) Use polar coordinates to find the area of $x^2 + y^2 \leq R^2$, where $R > 0$.
- (b) Under $x = 2u, y = 3v$, find $|J_T|$. Map $[0, 1]^2$ into the (x, y) -plane and verify its area by a change of variables.

Polar area and linear rescaling

$$\text{area}\{x^2 + y^2 \leq R^2\} = \int_0^{2\pi} \int_0^R r \, dr \, d\theta = \pi R^2.$$

For $T(u, v) = (2u, 3v)$,

$$|J_T| = \left| \det \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \right| = 6.$$

The unit square maps to $[0, 2] \times [0, 3]$, and

$$\int_0^1 \int_0^1 6 \, du \, dv = 6,$$

which is the rectangle's area.

Section 7

Applications

Which earlier idea is being used?

Each example below reuses one of the structures already developed.

Marginal $\rightarrow$ total FTC II accumulates a marginal quantity over an interval.

Surplus A definite integral measures area under inverse demand or supply.

Tail demand $\int_p^\infty f(v) dv$ is a tail accumulation; Leibniz differentiates the moving lower bound.

Present value An infinite horizon produces an improper integral; convergence becomes an economic condition.

Tail moments Expectations such as $\int xf(x) dx$ combine density weighting with improper-integral convergence tests.

Consumer and producer surplus

Consumer and producer surplus

With inverse demand $P_D(q)$, inverse supply $P_S(q)$, equilibrium (q^*, p^*) , and $P_D(q) \geq P_S(q)$ on $[0, q^*]$,

$$CS = \int_0^{q^*} P_D(q) dq - p^* q^*, \quad PS = p^* q^* - \int_0^{q^*} P_S(q) dq.$$

Consumer surplus: willingness to pay above what buyers pay.

Producer surplus: revenue above willingness to accept.

Total surplus is the area between demand and supply up to q^* .

Video: [Justin Eloriaga, Consumer, producer and social surplus using integration.](#)

Consumer and producer surplus

Consumer and producer surplus: figure

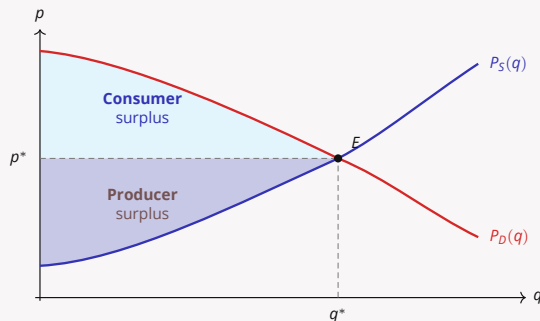


Figure 7 · Demand, supply, and the equilibrium price.

The equilibrium price partitions total surplus into consumer surplus and producer surplus.

Aggregate demand from valuations

Differentiating aggregate demand

Consumers have valuations v with density $f(v)$ and cdf $F(v)$. At price p , buyers are those with $v \geq p$:

$$D(p) = \int_p^{\infty} f(v) dv = 1 - F(p).$$

At prices p where the density f is continuous, Leibniz rule gives

$$D'(p) = -f(p).$$

The demand slope at p is minus the density of consumers who are just indifferent at that price.

Present value and growth

Infinite-horizon present value

If $u_0 > 0$, flow payoff grows at rate g , and discounting is at rate r ,

$$PV = \int_0^{\infty} e^{-rt} u_0 e^{gt} dt = u_0 \int_0^{\infty} e^{-(r-g)t} dt.$$

$$PV = \begin{cases} \frac{u_0}{r-g}, & r > g, \\ \text{diverges}, & r \leq g. \end{cases}$$

Infinite-horizon objectives need discounting to dominate long-run growth.

Pareto tail mean

Pareto tails

For a Pareto tail with $x_m > 0$ and $\alpha > 0$,

$$f(x) = \alpha x_m^\alpha x^{-\alpha-1}, \quad x \geq x_m.$$

The mean is

$$E[X] = \int_{x_m}^{\infty} x f(x) dx = \alpha x_m^\alpha \int_{x_m}^{\infty} x^{-\alpha} dx.$$

This integral is finite iff $\alpha > 1$. Heavy-tailed economic variables can have finite medians but infinite means.

Surplus, present value, and Pareto moments

- (a) Let $P_D(q) = 10 - q$ and $P_S(q) = 2 + q$. Find the market equilibrium and compute consumer, producer, and total surplus.
- (b) Evaluate $\int_0^\infty e^{-rt} e^{gt} dt$ and state when it converges.
- (c) For $f(x) = \alpha x_m^\alpha x^{-\alpha-1}$, $x \geq x_m > 0$, determine when $E[X]$ and $E[X^2]$ are finite and compute them when they are finite.

Surplus and present value

Equating demand and supply gives $q^* = 4$ and $p^* = 6$. Hence

$$CS = \int_0^4 (10 - q - 6) dq = 8, \quad PS = \int_0^4 (6 - 2 - q) dq = 8,$$

so total surplus is 16. For the present value,

$$\int_0^\infty e^{-(r-g)t} dt = \frac{1}{r-g}$$

when $r > g$; the integral diverges when $r \leq g$.

Pareto moments

For $k > 0$,

$$E[X^k] = \alpha X_m^\alpha \int_{X_m}^{\infty} x^{k-\alpha-1} dx$$

is finite when $\alpha > k$. Therefore

$$E[X] = \frac{\alpha X_m}{\alpha - 1} \quad (\alpha > 1), \quad E[X^2] = \frac{\alpha X_m^2}{\alpha - 2} \quad (\alpha > 2).$$